Online Supplementary Materials:

Experimental Instructions for "Leaders as role models and 'belief managers' in social dilemmas", *Journal of Economic Behavior & Organization* 2018

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The instructions were originally written in German. We report the instructions of the leader treatment.

General Information on the Experiment

You are taking part in an experiment on decision-making. If you read the following instructions carefully, you can, depending on your decisions, earn a considerable amount of money. It is therefore very important that you understand the following instructions.

- **Earnings**

In the experiment you earn points. The points you have earned will be converted in Euros at the rate

1 point = 0.02 €.

At the end of the experiment your total income in Euros will be paid to you in cash.

- **Group membership and anonymity**

During the experiment you are member of a group of four participants, i.e. there will be three more members in your group. The composition of the group will be the same during the whole experiment. Thus you form a group with the same participants throughout the experiment. It will be randomly determined with whom of the other participants you will form a group.

All participants decide anonymously, i.e. no participant will ever learn the identity of the other members of his group. To ensure anonymity it is imperative that all participants observe the following rule:
During the experiment all communication is prohibited, i.e. you are not allowed to speak or otherwise express yourself.
Should you have any questions please ask the experimenter.

The Experimental Procedures

The experiment consists of several periods.

The decision situation in period 1-10

The procedure is the same for each of these periods 1-10. Every participant receives **20 tokens** at the beginning of each period. Your task is to decide how you use your endowment. You have to decide how many of the 20 tokens you want to contribute to a project and how many of them to keep for yourself.

The calculation of your income

Your income consists of two parts:

1) The **tokens which you have kept for yourself** ("Income from tokens kept") whereby 1 token = 1 point.
2) The **income from the project**. This income is calculated as follows:
   Your income from the project = 0.4 x the total contribution of all 4 group members to the project

Your income in points is therefore:

(20 – your contribution to the project) + 0.4*(total contributions of all 4 group members to the project)

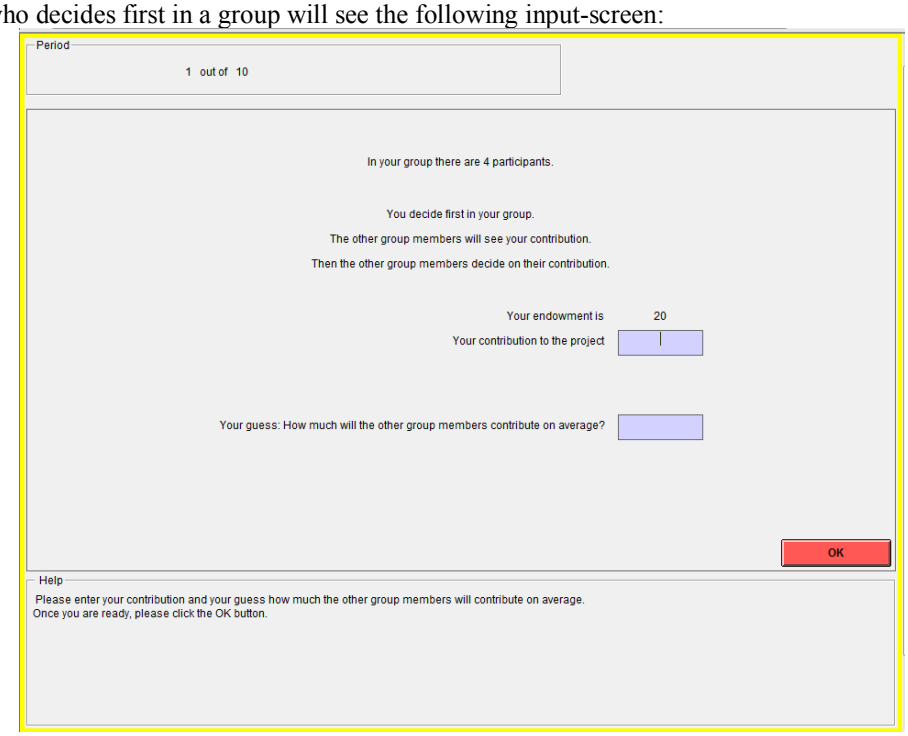
The income of each group member from the project is calculated in the same way, this means that each group member receives the same income from the project.

Examples:

Suppose the sum of the contributions of all group members is 60 tokens. In this case each member of the group receives an income from the project of: 0.4*60 = 24 points.
If the total contribution to the project is 9 tokens, then each member of the group receives an income of 0.4*9 = 3.6 points from the project.
For each token, which you keep for yourself you earn an income of 1 point. Suppose you contributed this token to the project instead, then the total contribution to the project would rise by one point. Your income from the project would rise by 0.4*1=0.4 points. However the income of the other group members would also rise by 0.4 points each, so that the total income of the group from the project would rise by 1.6 points. Your contribution to the project therefore also raises the income of the other group members. On the other hand you earn an income for each point contributed by the other members to the project. For each point contributed by any member you earn 0.4*1=0.4 points.

The decision

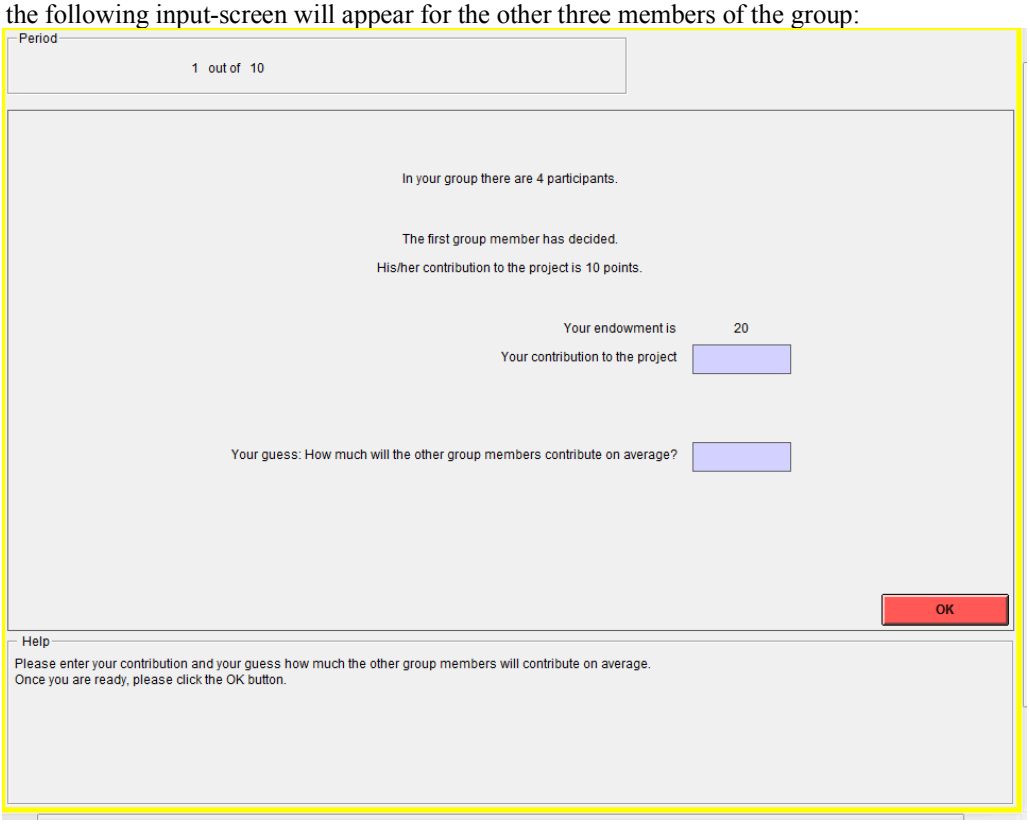
One member in each group **decides first** on his/her contribution. When this member has made his/her decision, the other members of the group are informed of the amount of tokens he/she has contributed to the project. Thereafter the other group members decide.
The person, who decides first in a group remains the same for all 10 periods.
The person who decides first in a group will see the following input-screen:



The period number appears in the top left of the screen. In the middle of the screen you will find the following information: Your group consists of 4 members**. You are the person who decides first. Your contribution will be displayed to the other members of your group. Then they will decide on their contribution. Your endowment is 20 tokens.**

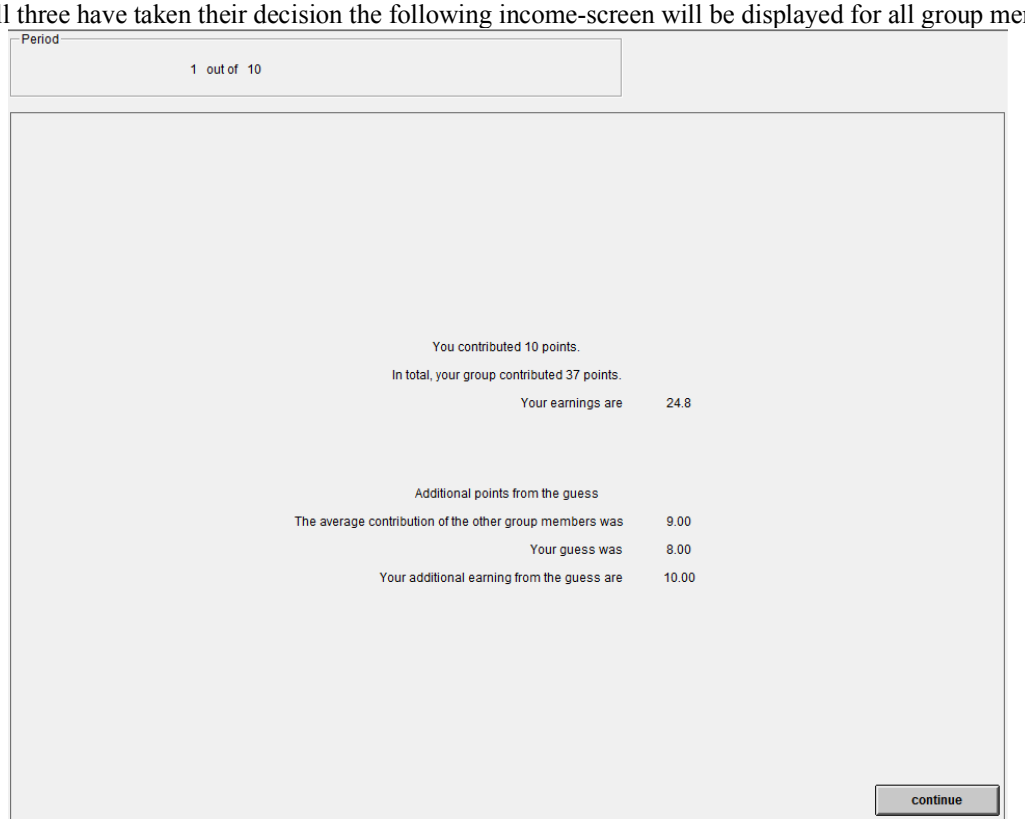
If this screen appears on your monitor you are the member of your group who decides first. You take your decision by typing a number between 0 and 20 in the input field. This field can be reached by clicking it with the mouse. As soon as you have decided how many points to contribute to the project, you have also decided how many points you keep for yourself: This is (20 – your contribution) tokens. After entering your contribution you must press the O.K. button (either with the mouse, or by pressing the Enter-key). Once you have done this, your decision can no longer be revised.

Below the decision field is another input field labeled “Your guess: How much will the other group members contribute on average”. Please type in what you think will be the average contribution of the other three members in your group. In addition to your earnings from your decision you will be paid an extra amount depending on how close your estimate is to the actual average contribution of the other group members. If your estimate is exactly right or not more than 0.5 points away from the actual average payoff you will earn 20 points. If your estimate is further off than 0.5 points you will earn 10 points divided by the (absolute) distance between your estimate and the actual average contribution.
Please click O.K. as soon as you have made your decision and typed in your estimate. After you have made the decision the following input-screen will appear for the other three members of the group:



The contribution of the group member, who has made his/ her contribution decision first will be displayed. You take your decision by typing a number between 0 and 20 in the input field and confirm by pressing the O.K.-button.
Please enter in the second input field labeled “Your guess:...” what you think is the average contribution of the other two group members. Here we ask you to estimate as accurately as possible what you think is the average contribution of the two group members that have not yet taken their decision. Please do not consider the decision of the group member that had decided first. In addition to your earnings from your decision you will be paid an extra amount depending on how close your estimate is to the actual average contribution of the two other group members. If your estimate is exactly right or not more than 0.5 points away from the actual average payoff you will earn 20 points. If your estimate is further off than 0.5 points you will earn 10 points divided by the (absolute) distance between your estimate and the actual average contribution.

After all three have taken their decision the following income-screen will be displayed for all group members:



Your own contribution and the contribution of the member who has made his/her decision first in your group are displayed. Furthermore, you are informed about the total contribution of all group members (including your contribution) and your income from your decision in the current period. Do you have any questions?